

Git for humans

Alice Bartlett

Senior Developer, Financial Times

@alicebartlett

HELLO

@alicebartlett

FINANCIAL TIMES

GDS



**Things are just *better*
when designers are
involved**

**“What tools could you not
live without that
designers don’t have?”**

Git

WHAT IS GIT

A man with short dark hair, wearing a dark t-shirt, is speaking at a white podium. He is gesturing with his hands, holding a small object in his right hand and pointing with his left. The background is a solid dark blue.

**“Git is an
application that
runs on your
computer, like a
web browser or a
word processor”**

Tom Stuart
<http://codon.com/>

WHAT DOES IT DO?

**Git helps you manage work
done on projects.**

**GIT IS
UNFRIENDLY**


```
o-techdocs — bash — 172x45
fsevent_watch ... bash ... bash bash bash node bash +
ft-origami o-footer o-techdocs strathausen-dracula-a6a5fa7
fticons o-forms o-typography test
google-amp o-ft-icons-blog-post o-video top_u_r_l_hits_20160205_150147.csv
headshot-images o-grid origami-build-service
logo-images o-header origami-build-tools
n-light-signup o-header-readme-draft origami-image-service

20:29:08-alice.bartlett~/Code$ git checkout o-techdocs/
fatal: Not a git repository (or any of the parent directories): .git
20:29:14-alice.bartlett~/Code$ cd o-techdocs/
20:29:18-alice.bartlett~/Code/o-techdocs (fix-code-color-contrast)$ git checkout master
Switched to branch 'master'
Your branch is up-to-date with 'origin/master'.
20:29:29-alice.bartlett~/Code/o-techdocs (master)$ git pull origin master
remote: Counting objects: 8, done.
remote: Total 8 (delta 4), reused 4 (delta 4), pack-reused 4
Unpacking objects: 100% (8/8), done.
From github.com:Financial-Times/o-techdocs
 * branch      master      -> FETCH_HEAD
    8e805e9..55e0b1e master  -> origin/master
Updating 8e805e9..55e0b1e
Fast-forward
 circle.yml   | 4 ++--
 origami.json | 1 +
 2 files changed, 3 insertions(+), 2 deletions(-)
20:29:39-alice.bartlett~/Code/o-techdocs (master)$ git branch
  add-pally
  fix-code-color-contrast
* master
  remove-benton
  removeBentonSans
20:29:54-alice.bartlett~/Code/o-techdocs (master)$ git branch -d add-pally
Deleted branch add-pally (was 6a139f6).
20:30:04-alice.bartlett~/Code/o-techdocs (master)$ git branch -d fix-code-color-contrast
Deleted branch fix-code-color-contrast (was 87fe768).
20:30:19-alice.bartlett~/Code/o-techdocs (master)$ git branch -d remove-benton
Deleted branch remove-benton (was 2e3cd0a).
20:30:29-alice.bartlett~/Code/o-techdocs (master)$ git branch -d removeBentonSans
Deleted branch removeBentonSans (was 8cf9a98).
20:30:39-alice.bartlett~/Code/o-techdocs (master)$ git branch
* master
20:30:42-alice.bartlett~/Code/o-techdocs (master)$ git checkout -b add-services-header
Switched to a new branch 'add-services-header'
20:30:52-alice.bartlett~/Code/o-techdocs (add-services-header)$ atom .
20:30:58-alice.bartlett~/Code/o-techdocs (add-services-header)$ atom .
20:33:00-alice.bartlett~/Code/o-techdocs (add-services-header)$
```

**There are other
applications you can use to
use Git.**

JARGON

GIT



**UNDERNEATH
ALL THIS, GIT IS
QUITE SIMPLE**

**WHY
ARE
YOU
HERE**

1. THING 1

2. THING 2

3. THING 3

4. THING 4

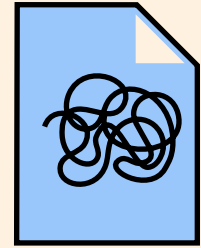
5. THING 5

THING 1:

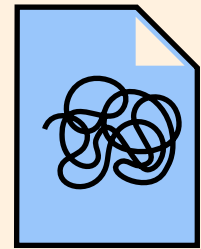
**GIT LETS YOU TELL
THE STORY OF YOUR
PROJECT**

**You use Git to take snapshots of
all the files in a folder.
This folder is called a repository
or repo.**

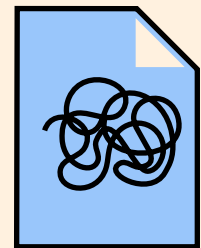
**When you want to take a
snapshot of a file or files, you
create a **commit****



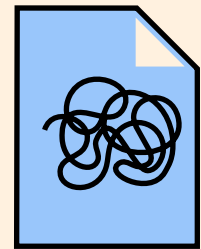
logo.svg



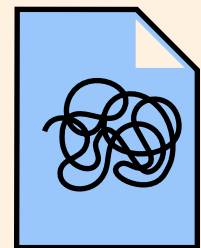
logo-2.svg



logo-3-monica-feedback.svg



logo-3-FINAL.svg

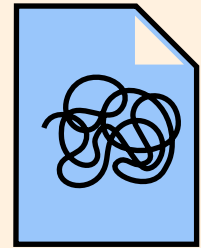


logo-3-FINAL-1.svg

By saving copies

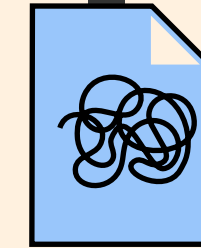
By making commits

By saving copies



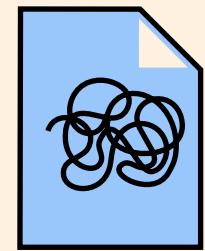
logo.svg

By making commits

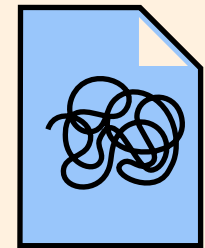


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By saving copies



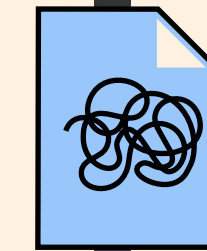
logo.svg



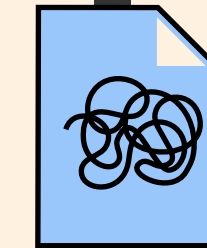
logo-2.svg



By making commits

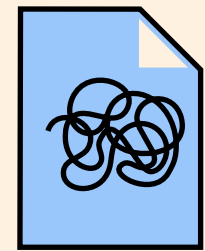


logo.svg

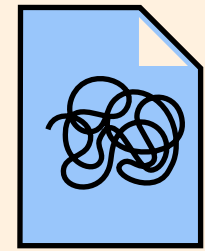


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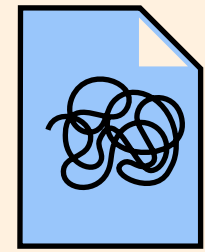
By saving copies



logo.svg



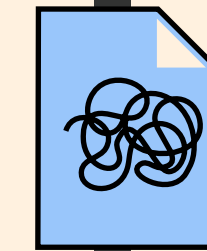
logo-2.svg



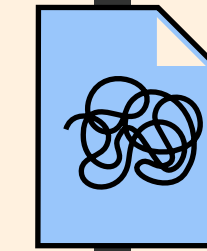
logo-3-monica-feedback.svg



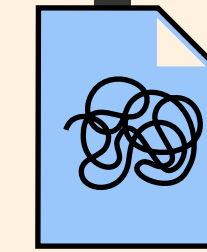
By making commits



logo.svg

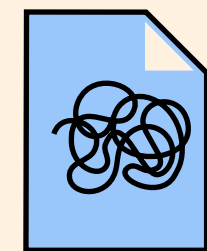


logo.svg

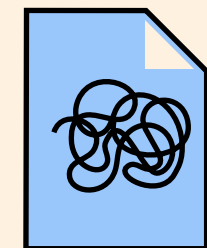


logo.svg

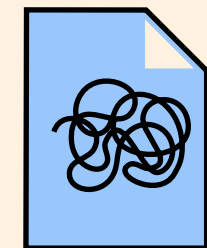
By saving copies



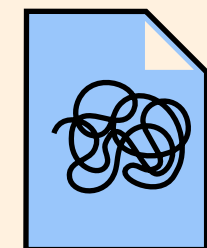
logo.svg



logo-2.svg



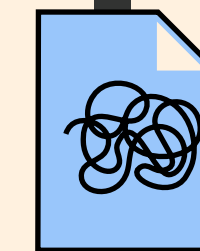
logo-3-monica-feedback.svg



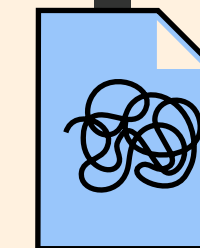
logo-3-FINAL.svg



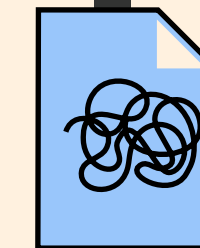
By making commits



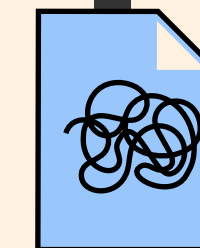
logo.svg



logo.svg

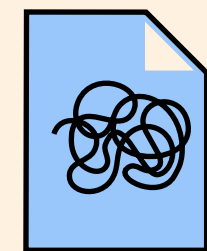


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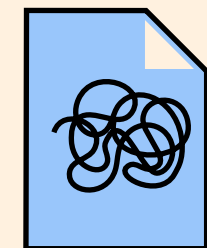


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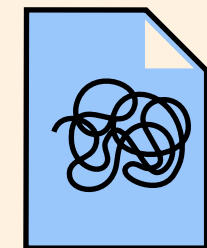
By saving copies



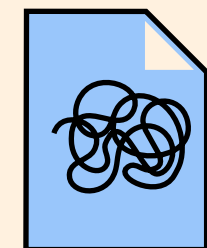
logo.svg



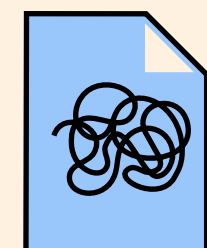
logo-2.svg



logo-3-monica-feedback.svg

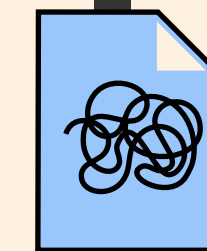


logo-3-FINAL.svg

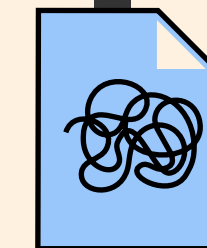


logo-3-FINAL-1.svg

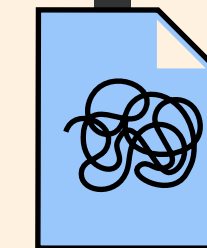
By making commits



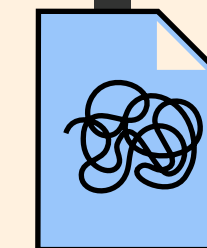
logo.svg



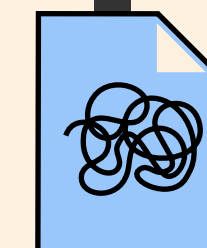
logo.svg



logo.svg



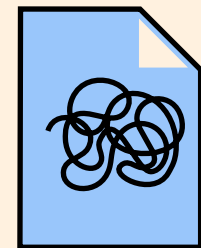
logo.svg



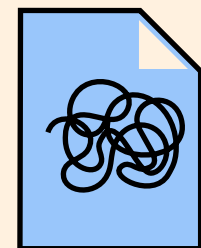
logo.svg



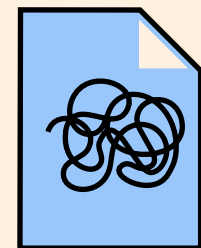
By saving copies



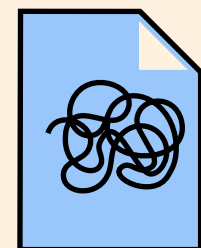
logo.svg



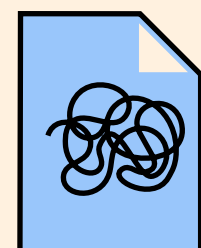
logo-2.svg



logo-3-monica-feedback.svg

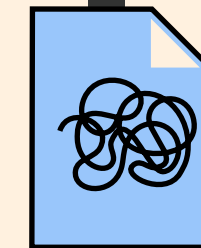


logo-3-FINAL.svg

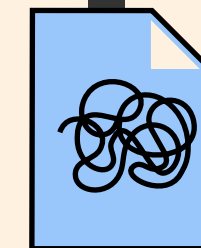


logo-3-FINAL-1.svg

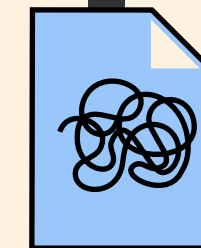
By making commits



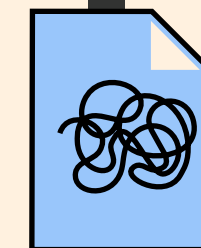
logo.svg



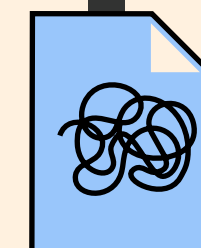
logo.svg



logo.svg



logo.svg



logo.svg

**When you `commit` a file or files,
some information is saved along
with the changes to the file**

1. Who

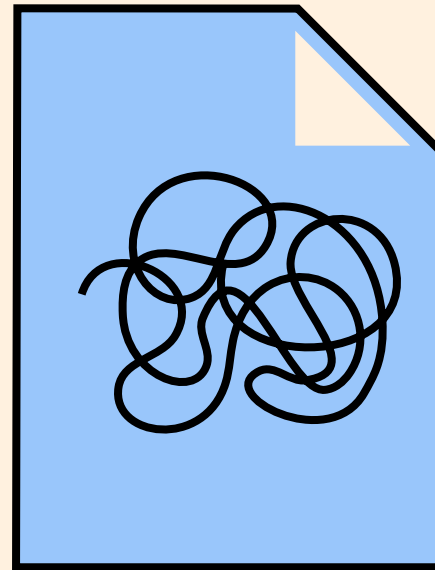
2. When

**You can add more information
about the changes you've made
in a **commit message****

A good commit message:

Update link style

User research showed that many people did not spot links in the copy. This commit updates the link style to the new underlined style which performed better.



logo-3-FINAL-1.svg

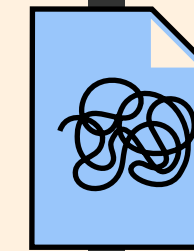
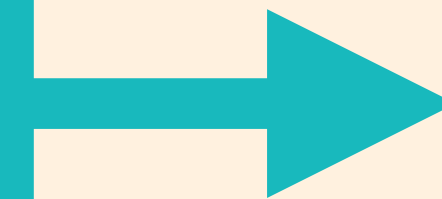
Alice Bartlett

10:34am March 11th 2016

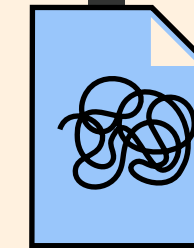
Update link style

User research showed that many people did not spot links in the copy. This commit updates the link style to the new underlined style which performed better.

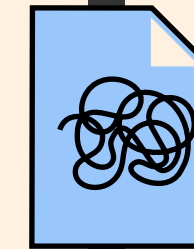
By making commits



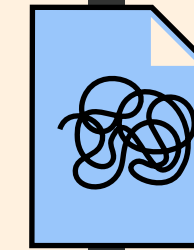
logo.svg



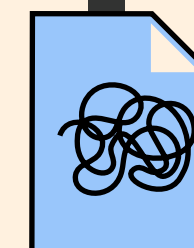
logo.svg



logo.svg



logo.svg



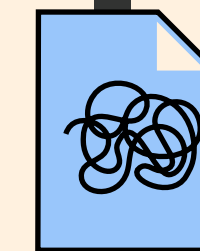
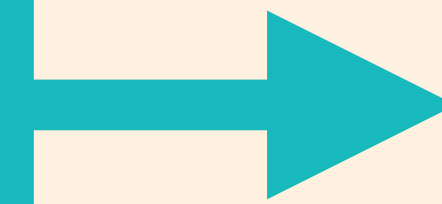
logo.svg

By making commits

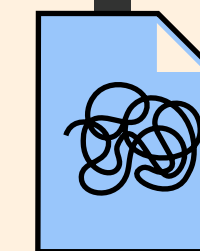
Alice Bartlett
12:43pm May 5th 2016

Add new colours

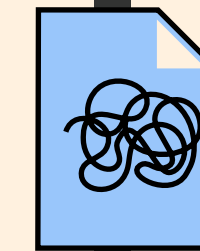
New colours for US election campaign



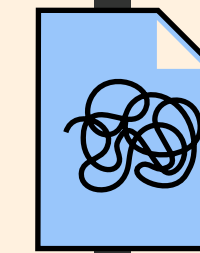
logo.svg



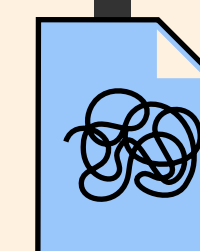
logo.svg



logo.svg



logo.svg



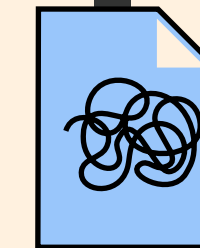
logo.svg

By making commits

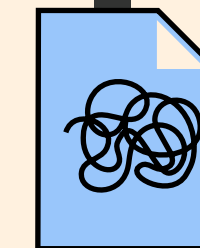
Alice Bartlett
12:43pm May 8th 2016

Fix Orange

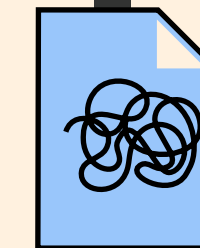
The orange we used fails AAA
accessibility contrast tests so beef it up
to contrast properly



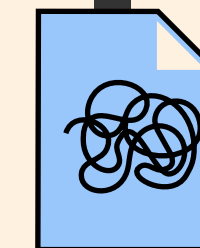
logo.svg



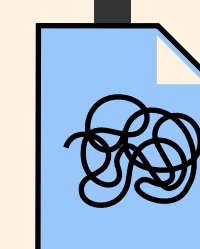
logo.svg



logo.svg

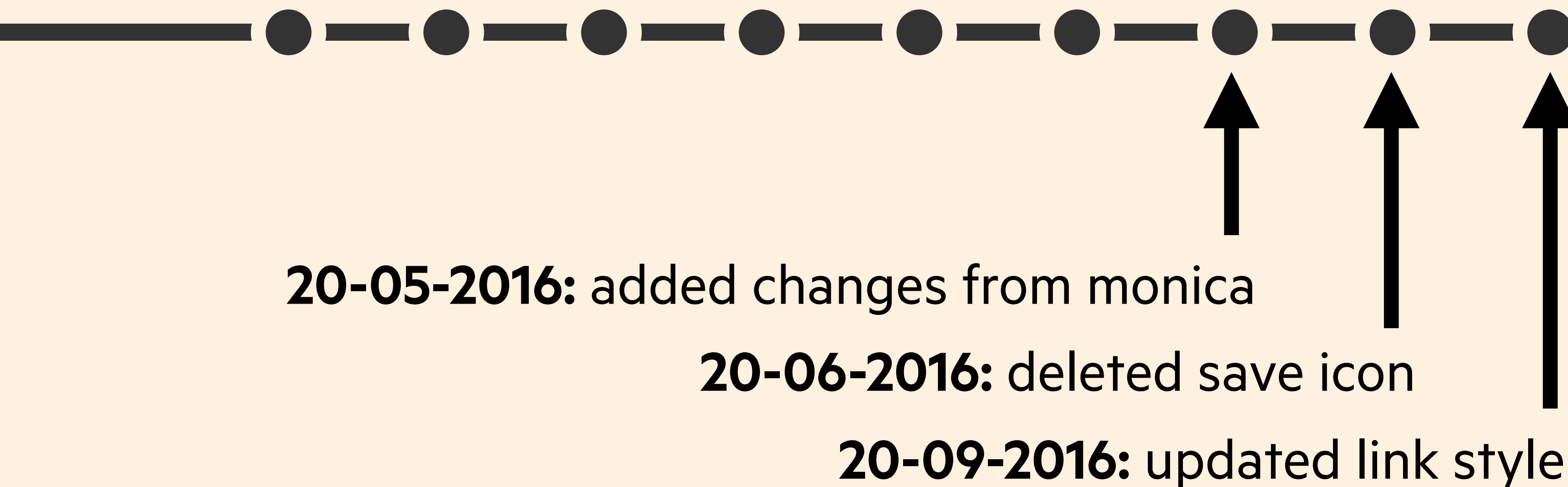


logo.svg



logo.svg

Git stores the whole history of your project



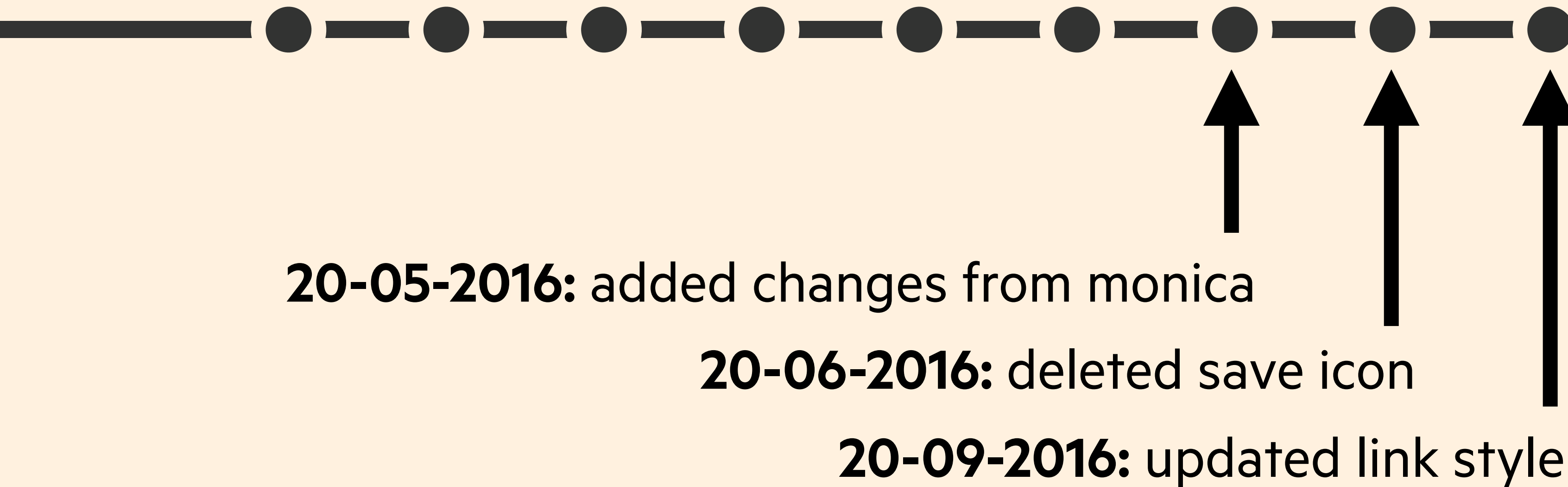
repository - your project folder
commit - save a snapshot

THING 2:

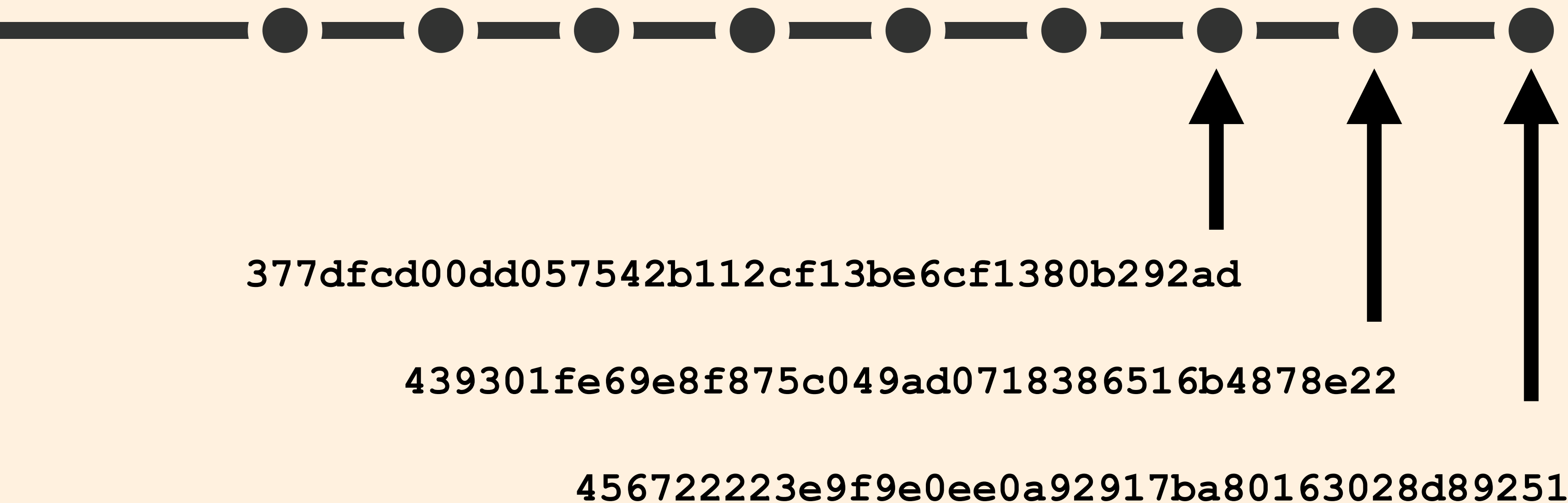
**GIT LETS YOU TIME
TRAVEL**

**Once you've saved some
snapshots, Git lets you move
through them**

Git stores the whole history of your project



Each of these commits has an id called a **hash**



I can tell Git what commit I want to check out using the commit hash



20-05-2016: deleted play icon

d5b87865bc2cd9d38ba8284c2eaa0d0241d800bb

**Getting the files from a commit
in the past is known as doing a
check out**

I can tell Git what commit I want to check out using the commit hash



20-05-2016: deleted play icon

d5b87865bc2cd9d38ba8284c2eaa0d0241d800bb

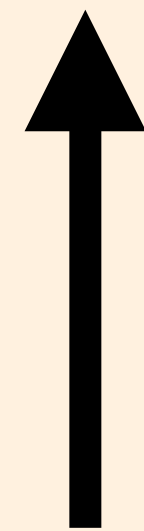
I can tell Git what commit I want to check out using the commit hash



20-05-2016: deleted play icon

d5b87865bc2cd9d38ba8284c2eaa0d0241d800bb

My other commits still exist, but when I look in my repo, it's as if they never happened



20-05-2016: deleted play icon

d5b87865bc2cd9d38ba8284c2eaa0d0241d800bb

hash - a computer generated id

checkout - time travel to a specific commit

THING 3:

**GIT HELPS YOU
EXPERIMENT**

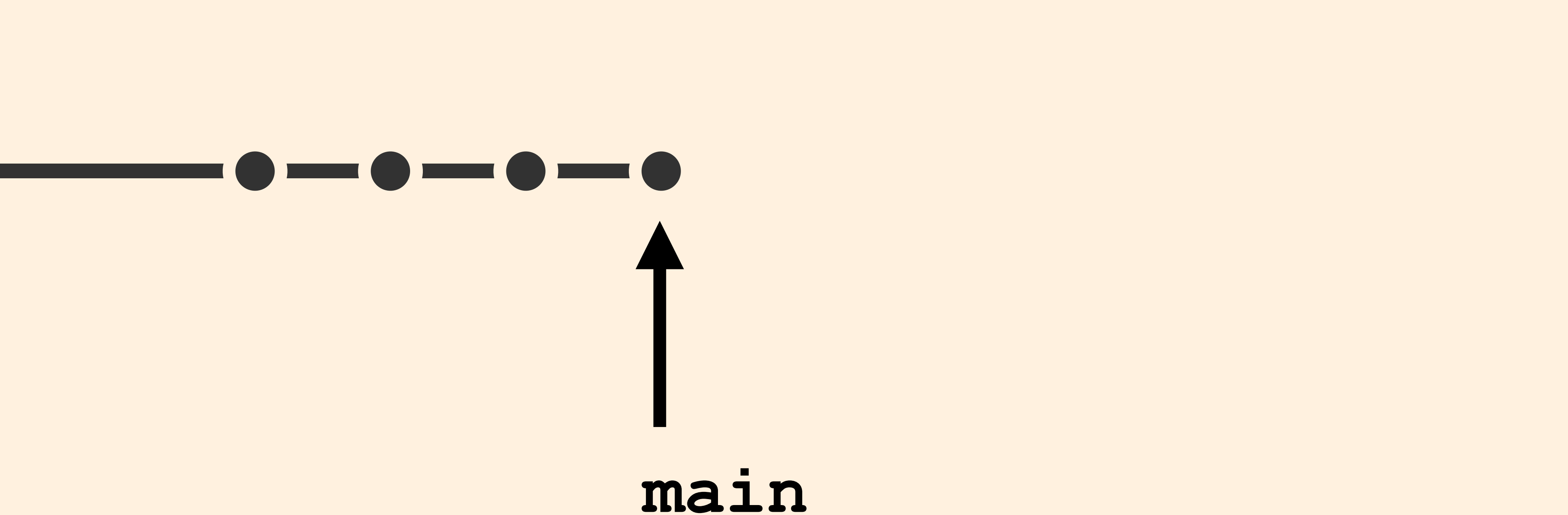
**So far, everything has been
very linear and ordered.**

**This isn't really how projects
work, sometimes you want to
make easily discardable
experiments**

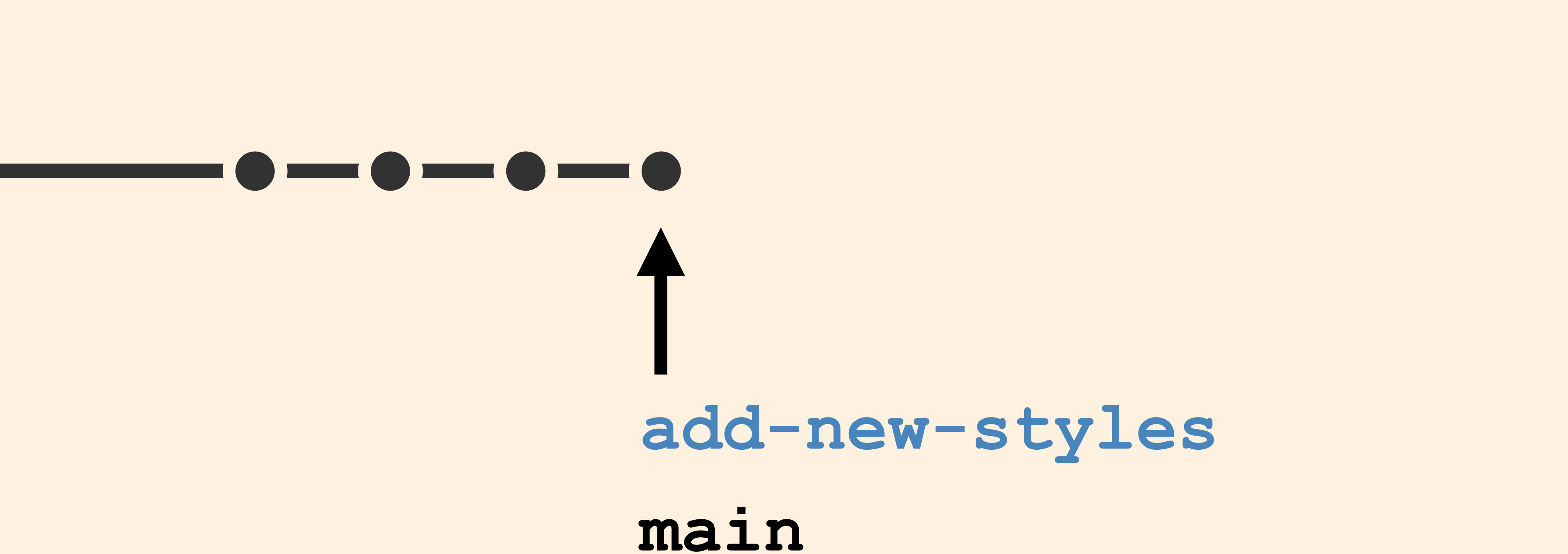
**The way you do this in Git is
with branches**

A **branch is a moveable label
attached to a commit**

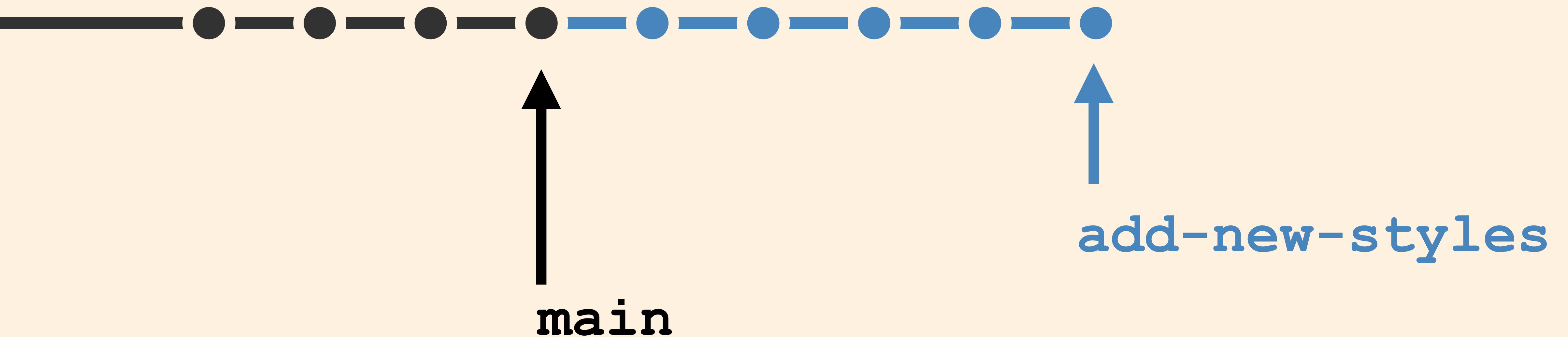
The default branch name in Git is **main**

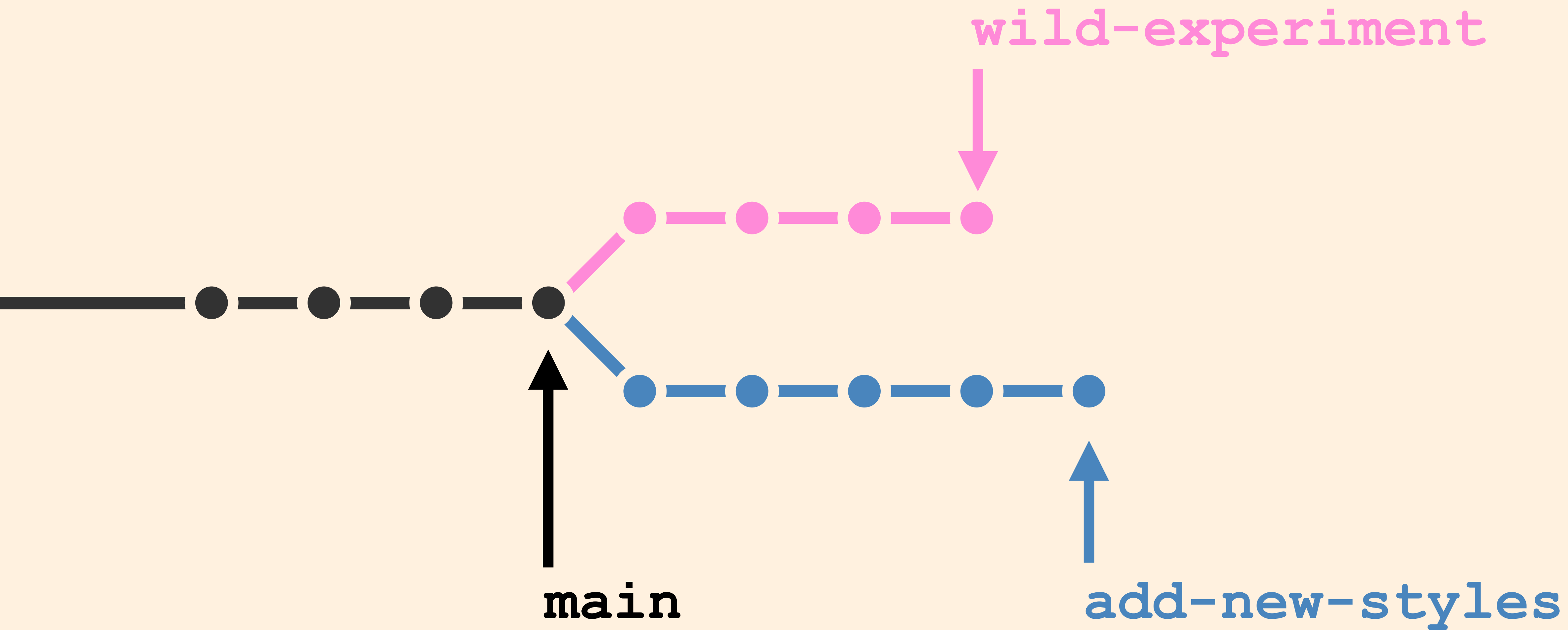


You can add your own branches too



A developer will often do lots of work on a branch





Branches are useful for trying out stuff, as they're really easy to throw away if you decide you don't like your changes

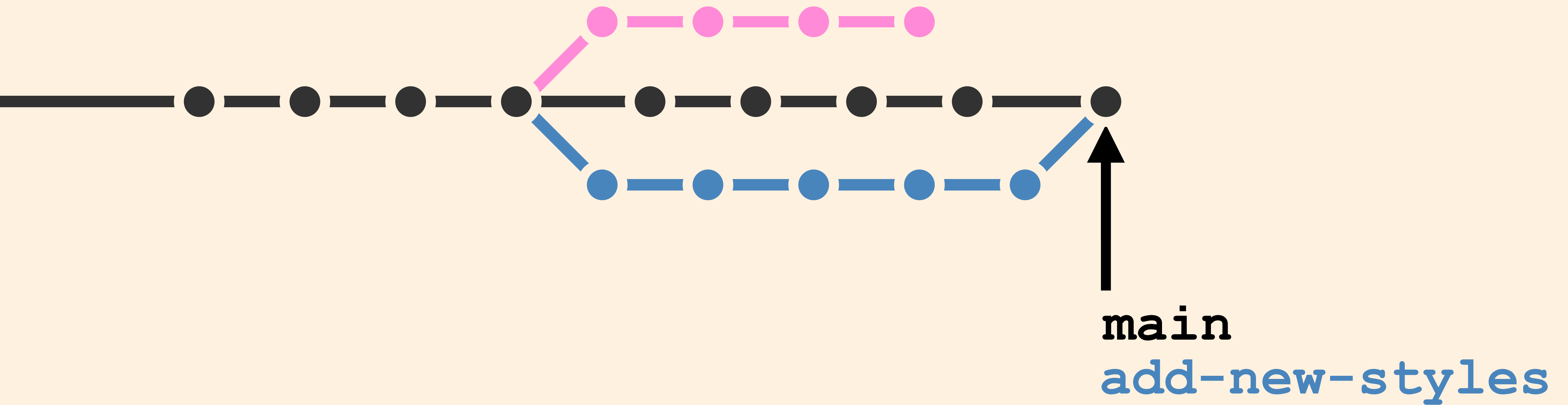
**At the FT most dev work is
done in branches, the main
branch is considered special**

**It's common for the main
branch to be the version of the
code or files that are live on the
site**

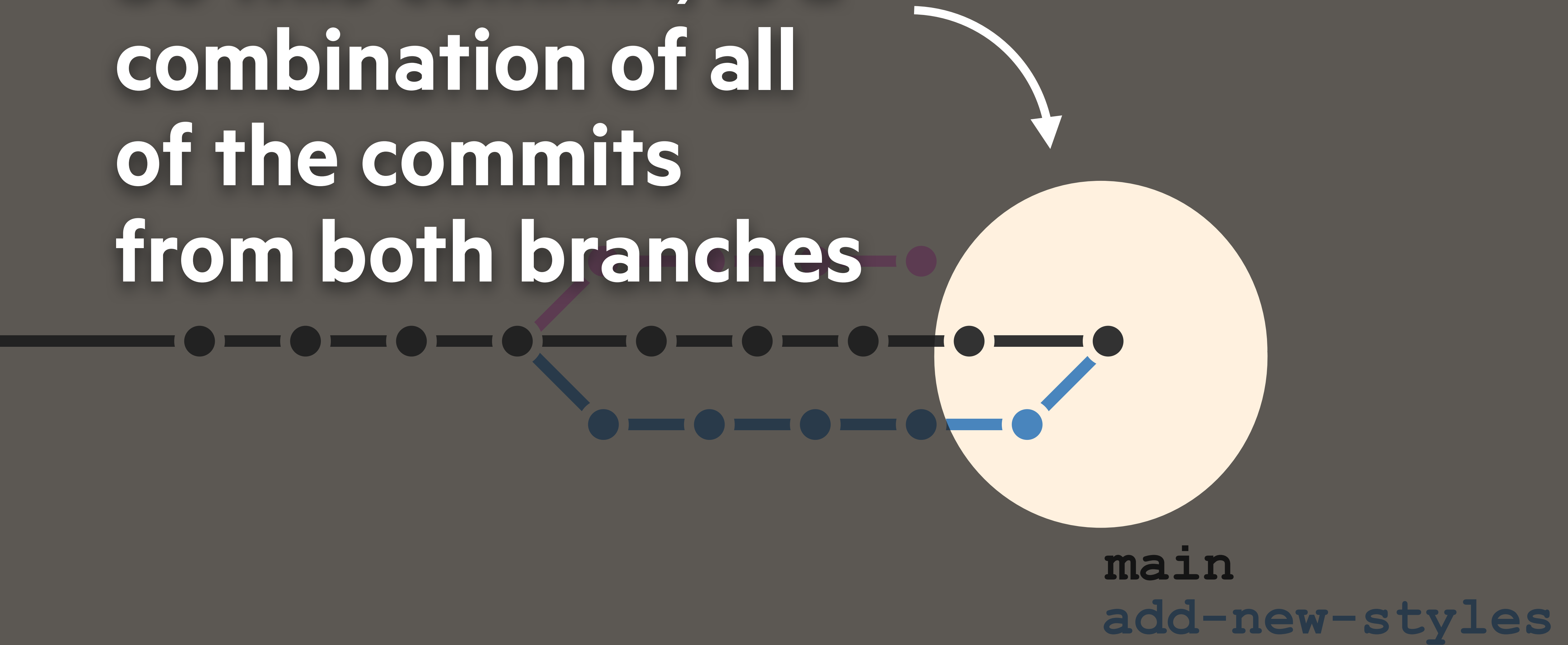
**Whereas other branches can
contain work in progress**

Once you're happy with some work, you need a way to get it back into main

To get changes from one branch into another, you **merge** them



So this commit, is a
combination of all
of the commits
from both branches



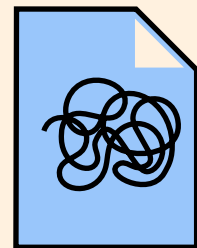
branch - a moveable label that points to a commit
merge - the combination of two or more branches

THING 4:

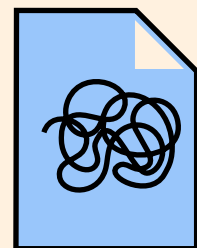
**GIT HELPS YOU BACK
UP YOUR WORK**

**Everyone knows that you should
back up your work regularly**

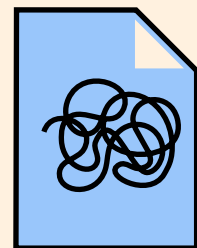
**Ideally to somewhere that is
geographically distinct from your
computer**



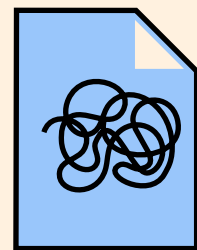
logo.svg



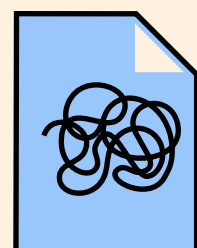
logo-2.svg



logo-3-monica-feedback.svg



logo-3-FINAL.svg

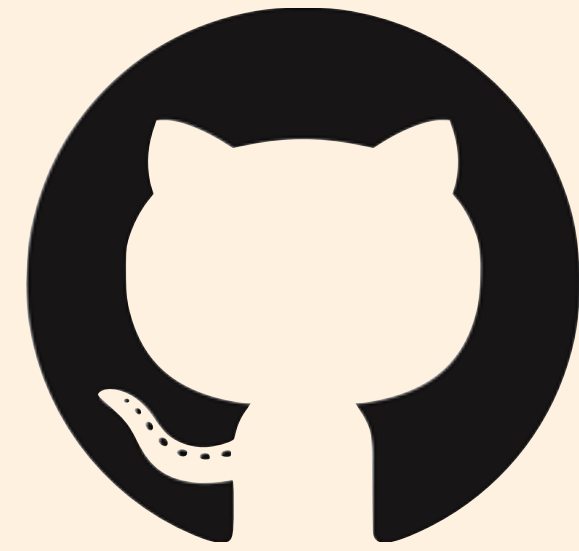


logo-3-FINAL-1.svg



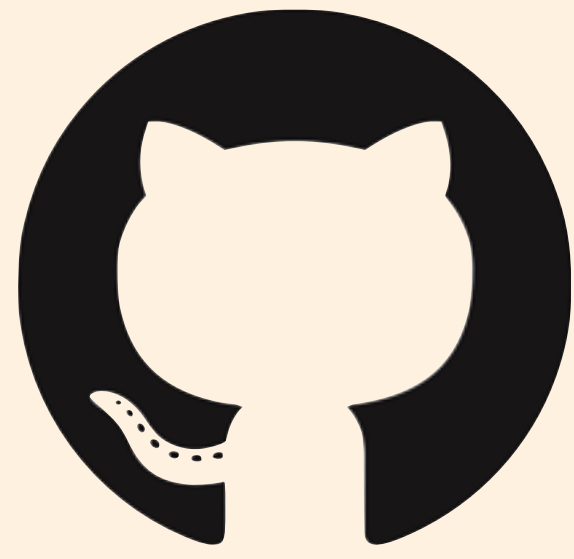
- **Safer**
- **Access from different places**
- **Shared access**

In Git this place is called a **remote**

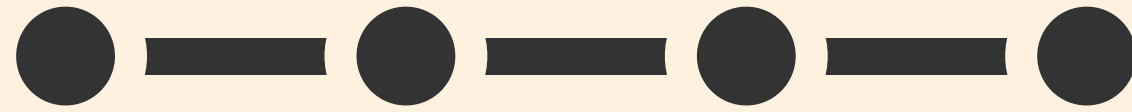


A very popular remote is Github

**To get some work from a remote
for the first time you clone it**



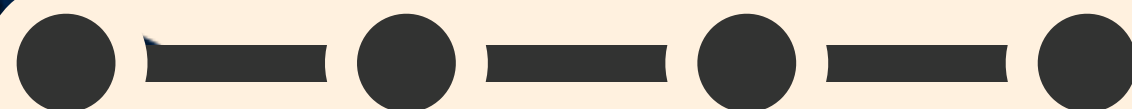
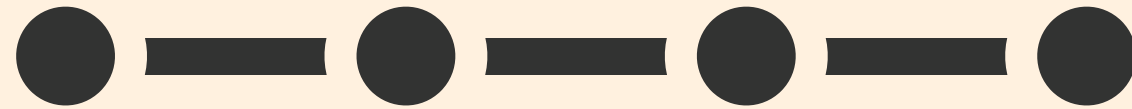
Remote



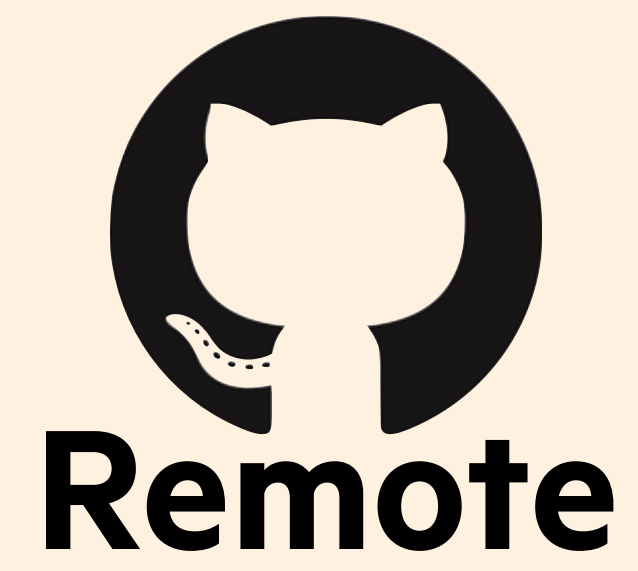
@alicebartlett



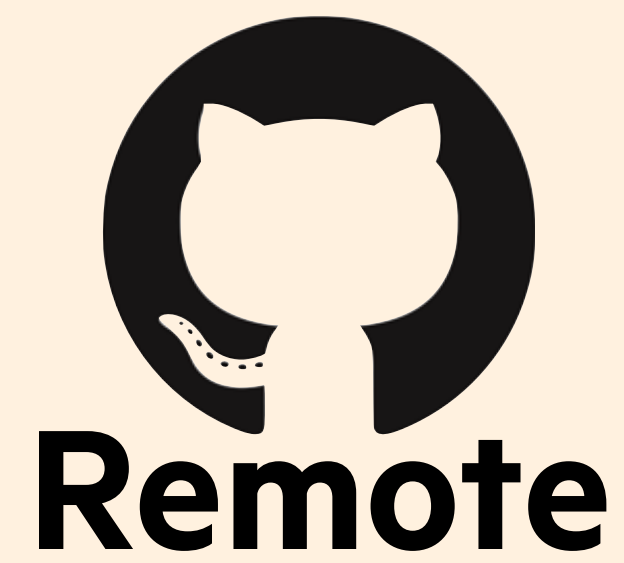
Remote



@alicebartlett

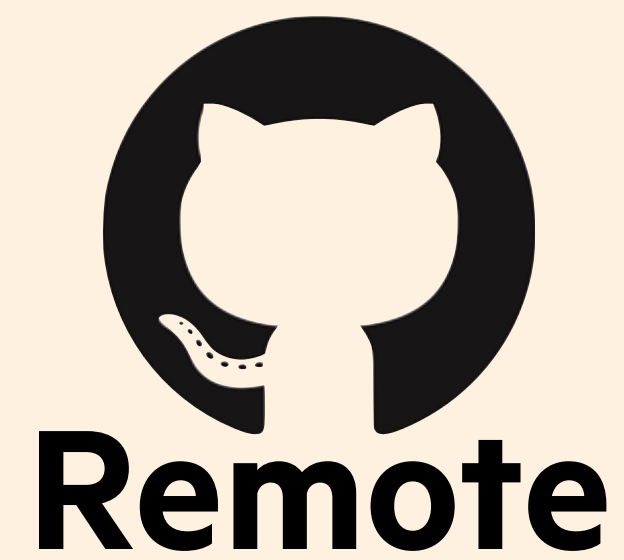


@alicebartlett



**Now everyone
has the repo on
their computer**

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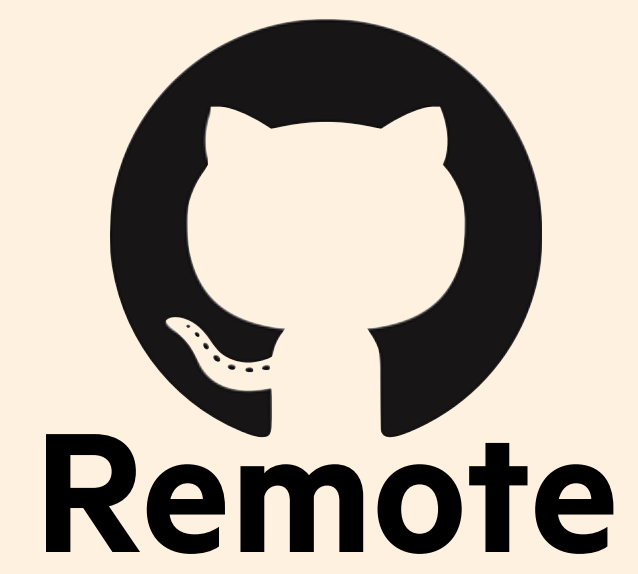
Lucy Kellaway
10:34am November 4th 2016

Fix broken icon tinting

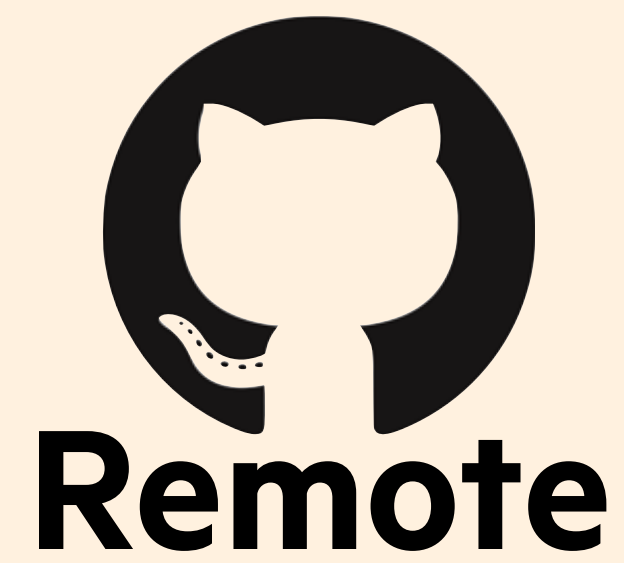
Icon tinting was case sensitive so `#FFF` worked but `#fff` didn't. This commit removes this bug.



@alicebartlett

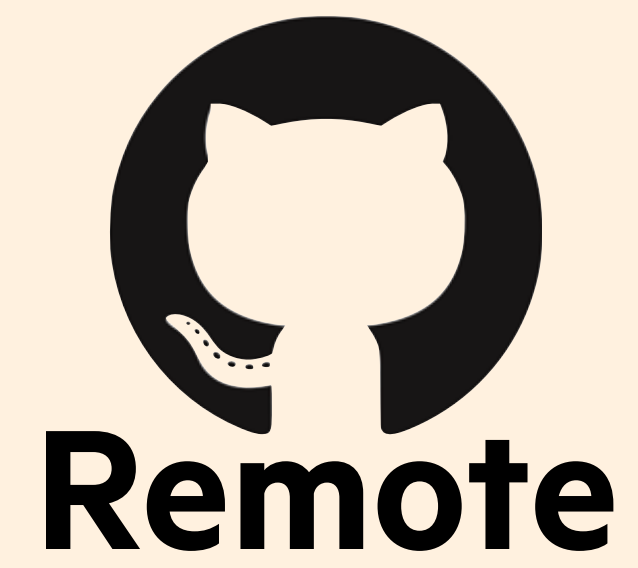


@alicebartlett

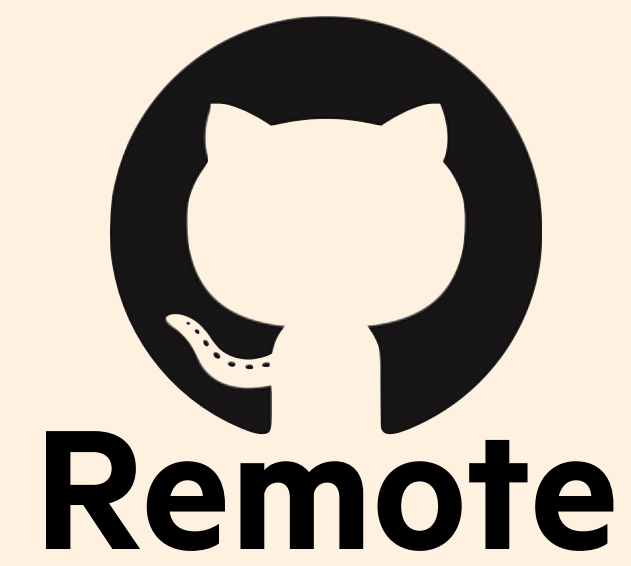


**Lucy can send
her changes to
remote**

@alicebartlett



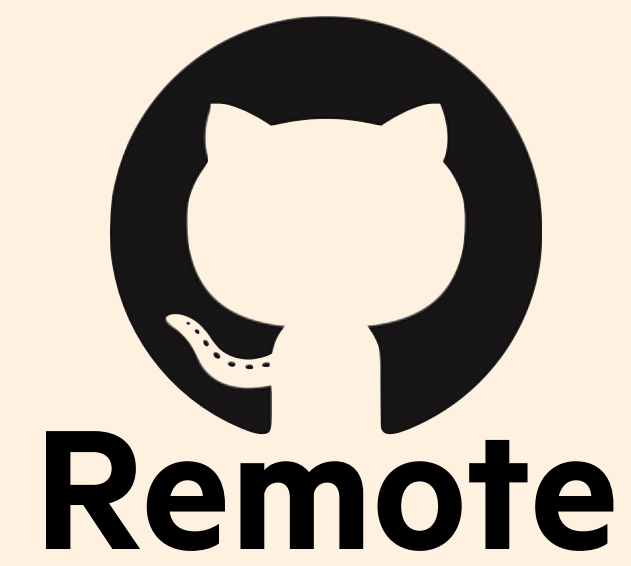
@alicebartlett



This is known as
a **push**

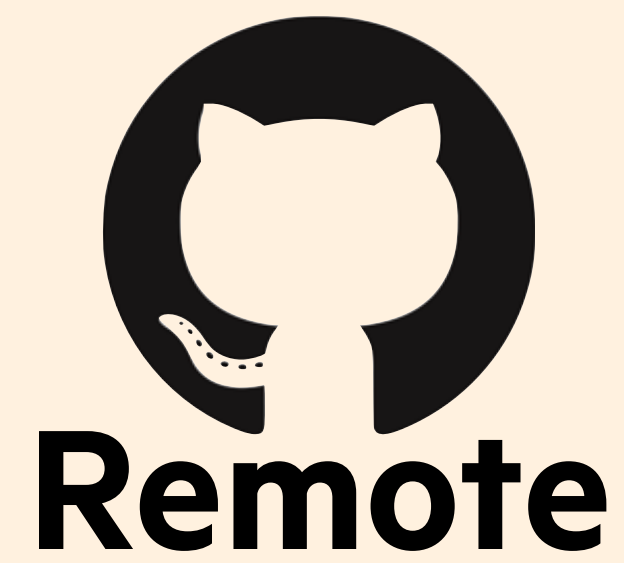


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**Now Martin is
behind**

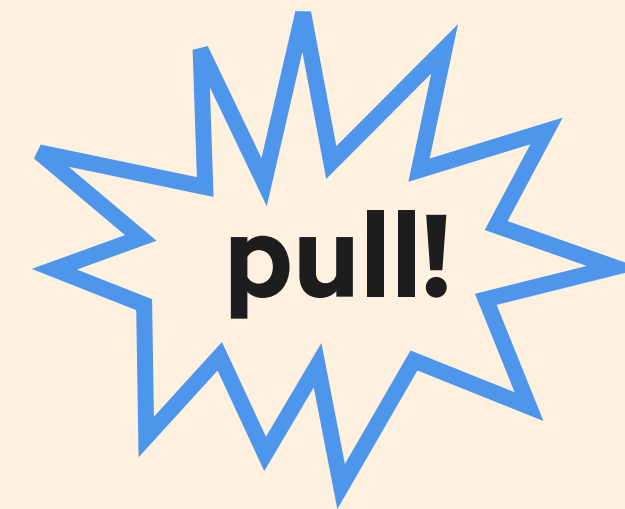
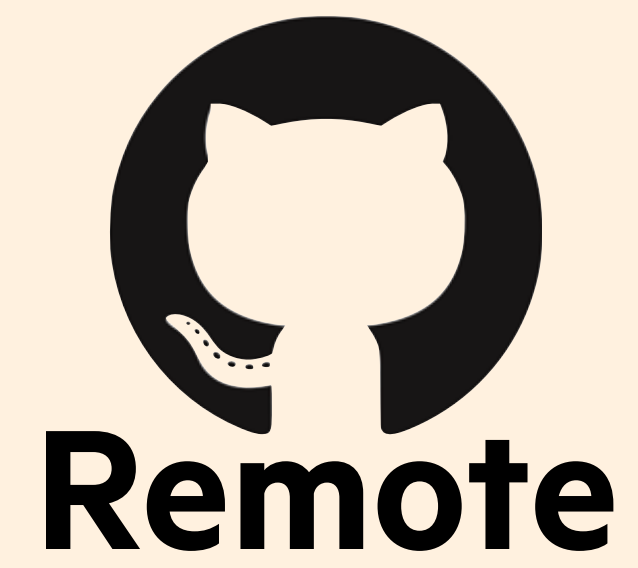
@alicebartlett



To get these
changes, Martin will
need to **pull** them



@alicebartlett



@alicebartlett

remote - a computer with a repo on it

clone - get the repo from the remote for the first time

pull - get new commits to the repo from the remote

push - send your new commits to the remote

THING 5:

**GIT HELPS YOU
COLLABORATE**

**Committing helps you tell other
people the story of your project**

**Remotes mean other people can
access your project**

**Merges help manage combining
your work with someone else's**

Git allows lots of people to work on the same project, which is why people suffer through the terrible UX of it.

Git terms we've covered

repository	your project folder
commit	a snapshot of your repo
hash	an id for a commit
checkout	time travel to a specific commit
branch	a movable label that points to a commit
merge	combining two branches
remote	a computer with the repository on it
clone	get the repository from the remote for the first time
push	send commits to a remote
pull	get commits from a remote

- 1. Tell the story of your project**
- 2. Travel back in time**
- 3. Experiment with changes**
- 4. Back up your work**
- 5. Collaborate on projects**

Thank you

Alice Bartlett

Senior Developer, Financial Times

@alicebartlett